

CO2xCOMM: Porewater Methane

July 2026 Samples

2026-08-28

Contents

1	Read in data files	2
2	Assess standard curves	2
3	Aanlayze the Check Standards	5
4	Analyze the Blanks	6
5	Dilution correct samples and Check Range	9
6	Calculate gas in water	10
7	Merge samples with Metadata & check all samples are present	10

##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

Run Information: Brenden D. Blakley

```
library(here)
```

here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
sample_month <- "July"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
GHG_files <- c("Raw Data/20260821_GCREW_2M_1M_formatted.csv")
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- here("Porewater QAQC Logs", "GCREW_CH4_QAQC_Log.csv")
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/GCREW_CO2xCOMM_Porewater_CH4_202607.csv"
```

```
#Gas in Water Calculations
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- ""
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

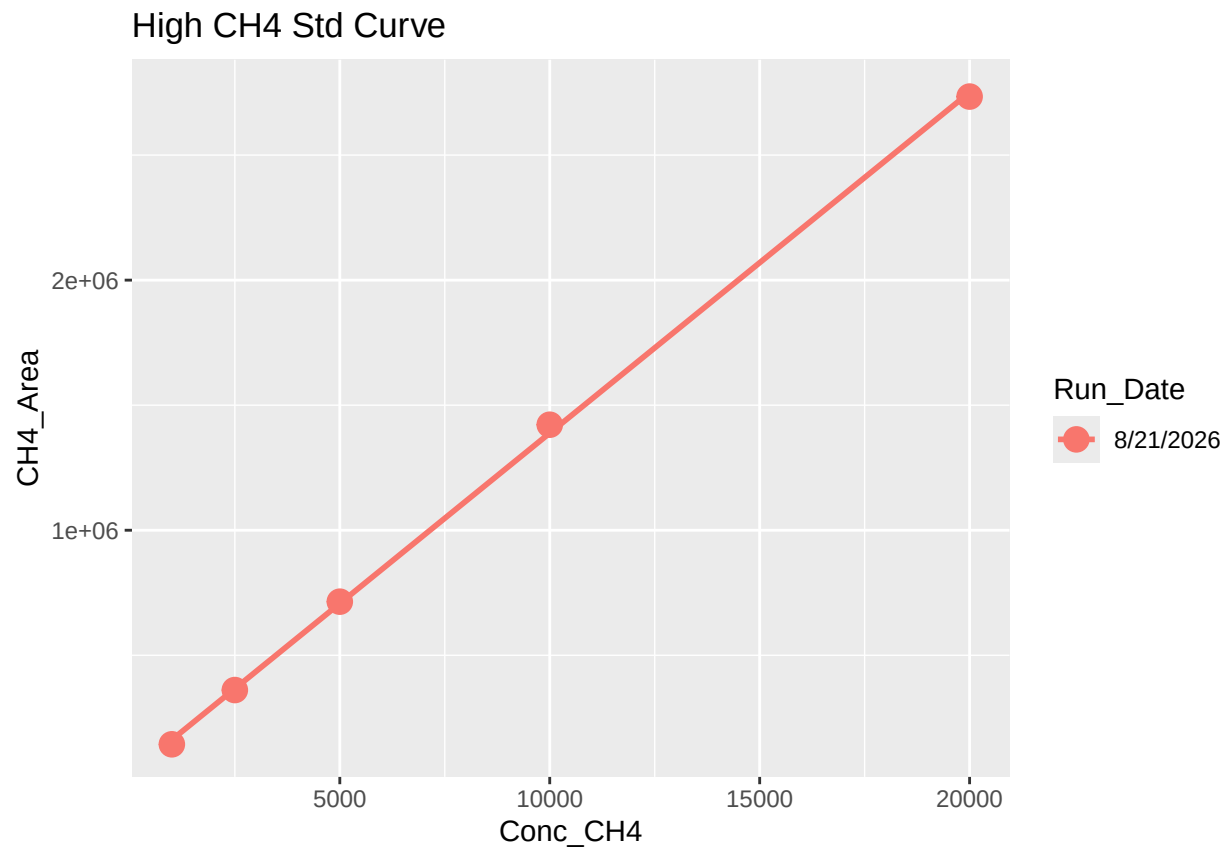
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

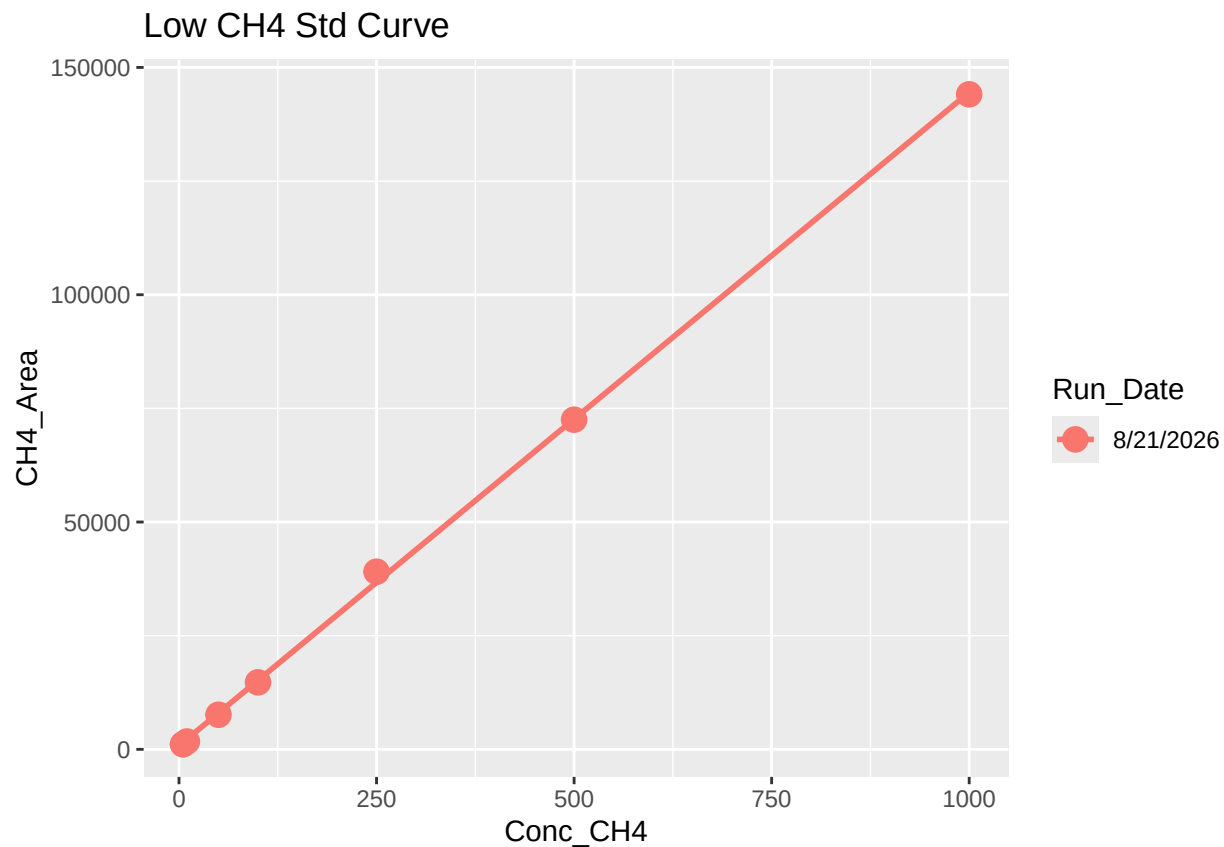
1 Read in data files

2 Assess standard curves

'geom_smooth()' using formula = 'y ~ x'



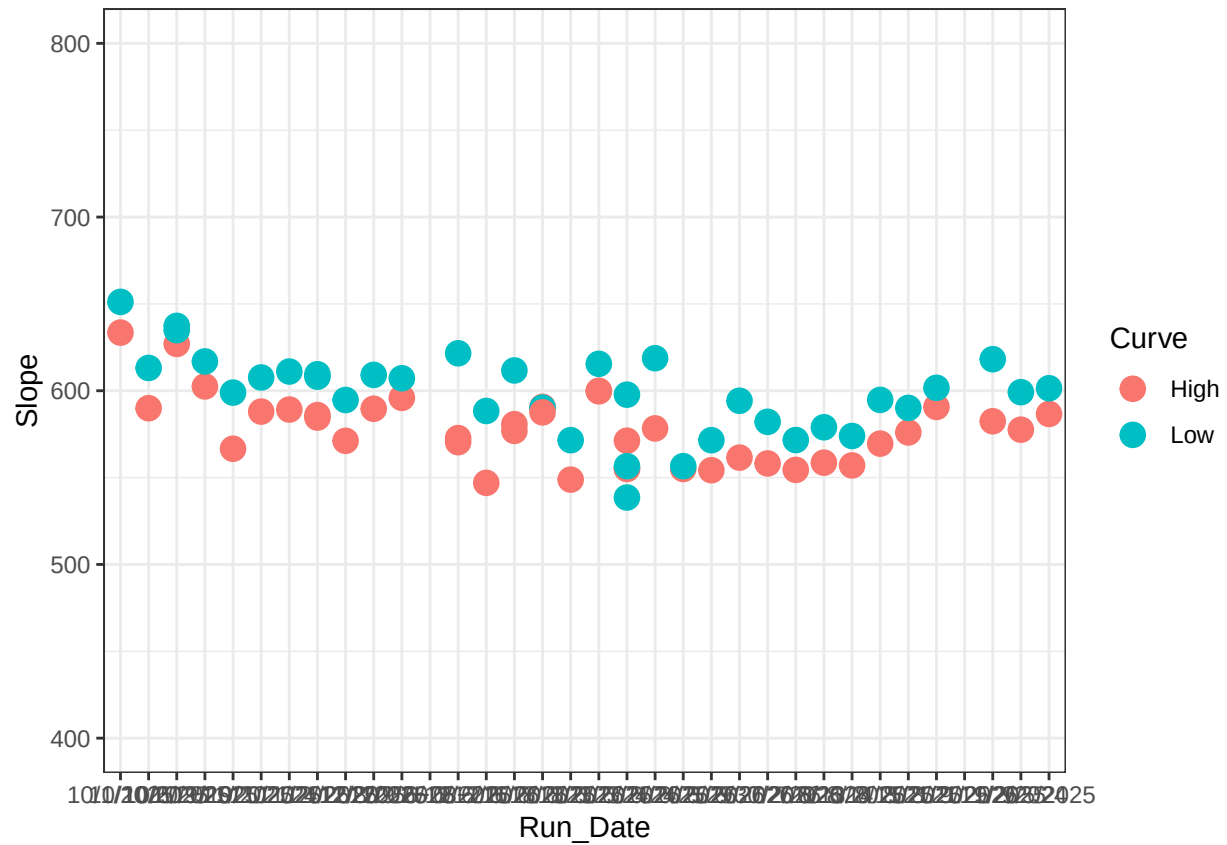
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'
```

```
##   Curve      R2    Slope  Intercept Run_Date
## 1  High 0.9986295 576.8990 173986.2174 6/16/2025
## 2  Low 0.9990260 611.6961  -3791.2684 6/16/2025
## 3  High 0.9999101 570.3925 138747.3478 6/12/2025
## 4  Low 0.9979098 621.5734   3224.2535 6/12/2025
## 5  High 0.9989882 590.7374 -109278.7391 6/18/2025
## 6  Low 0.9990995 590.3138    954.7682 6/18/2025
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```



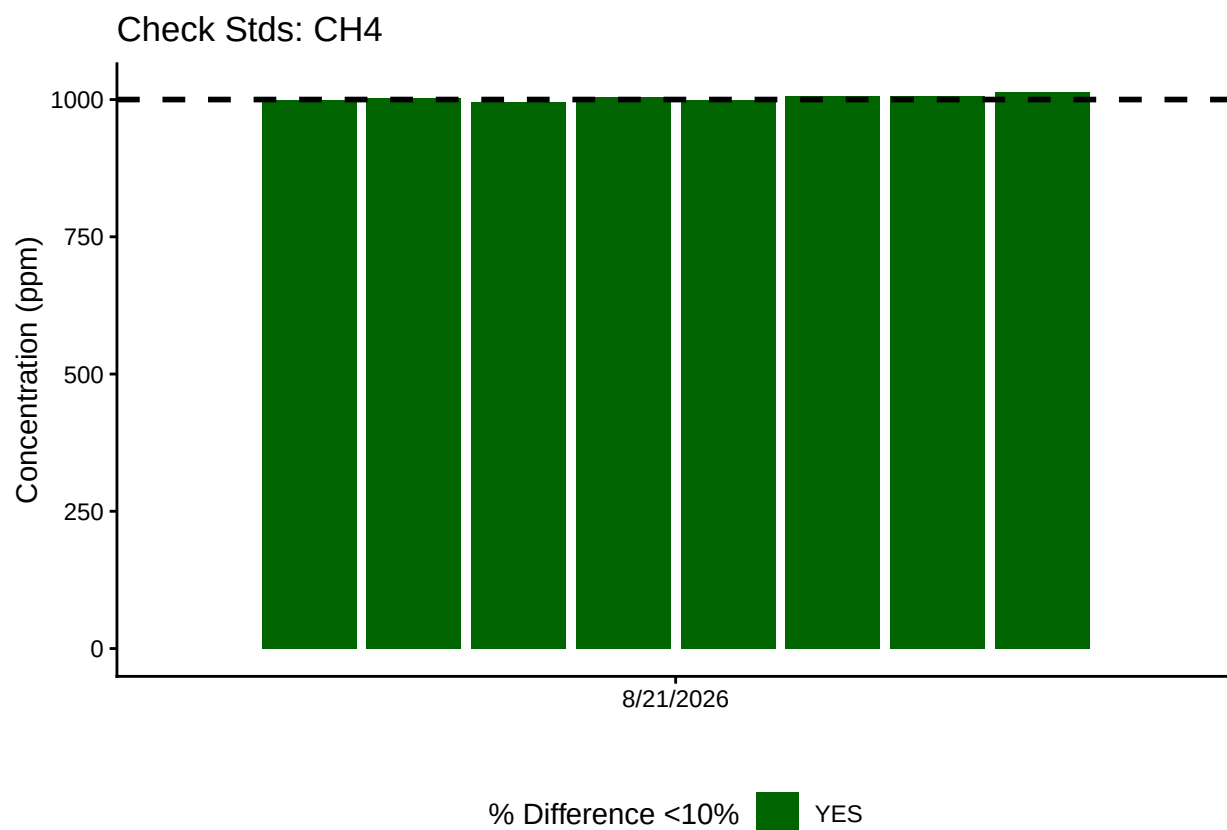
```
## [1] "High_R2 is Good-8/21/2026"
```

```
## [1] "Low_R2 is Good-8/21/2026"
```

2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

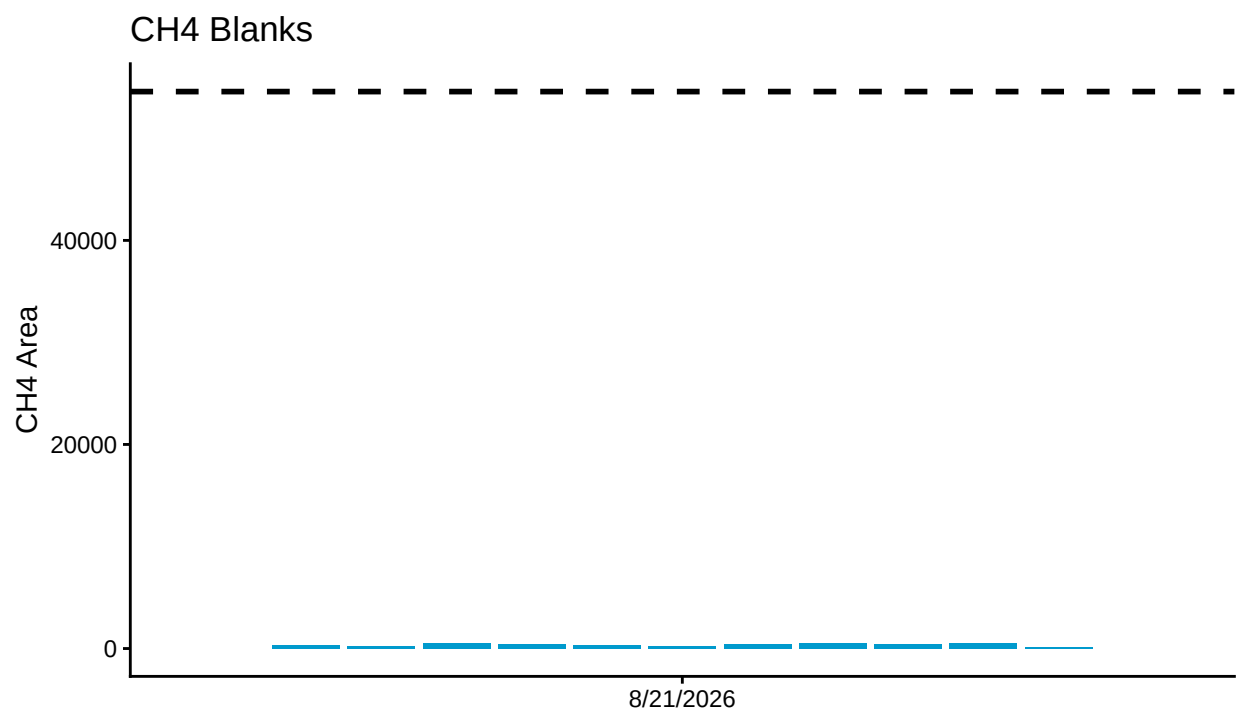
```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

Assess Blanks

>60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

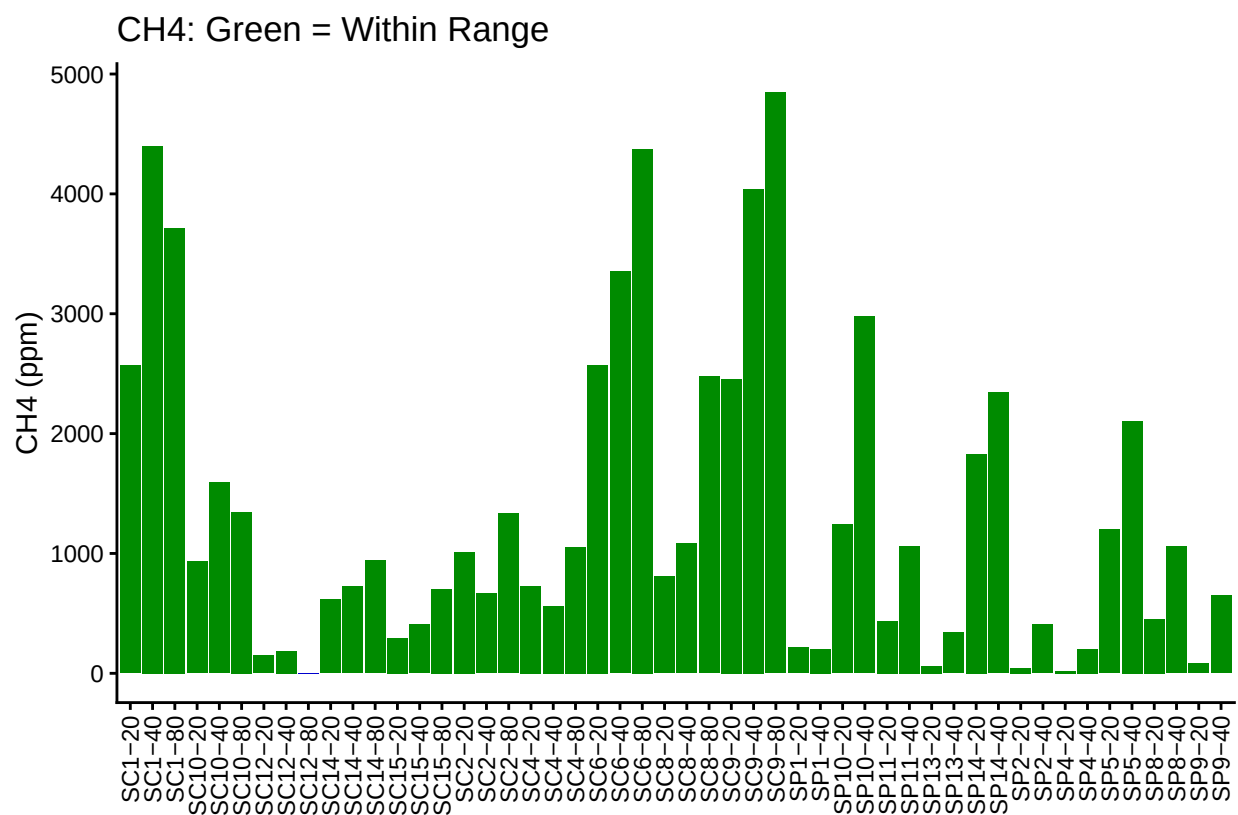
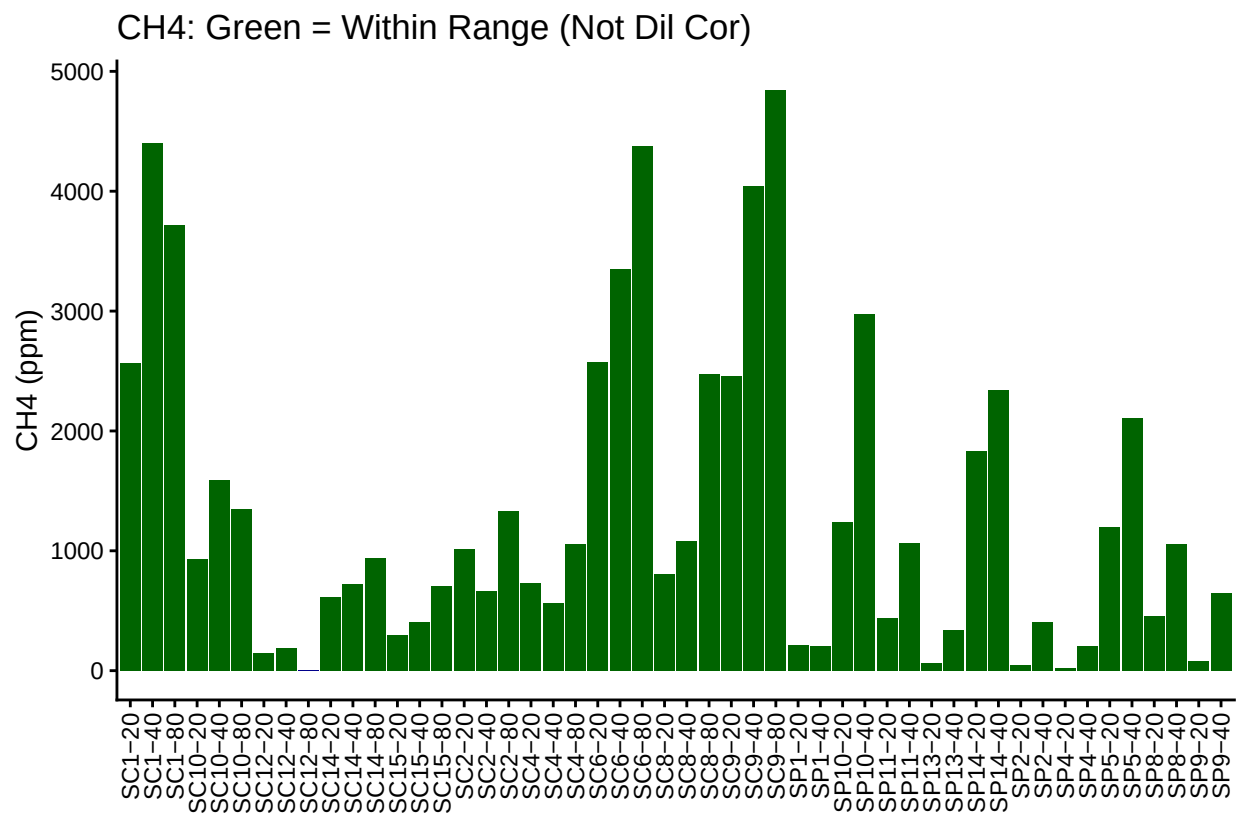


Blank Conc <25% Quartile Samples ■ YES

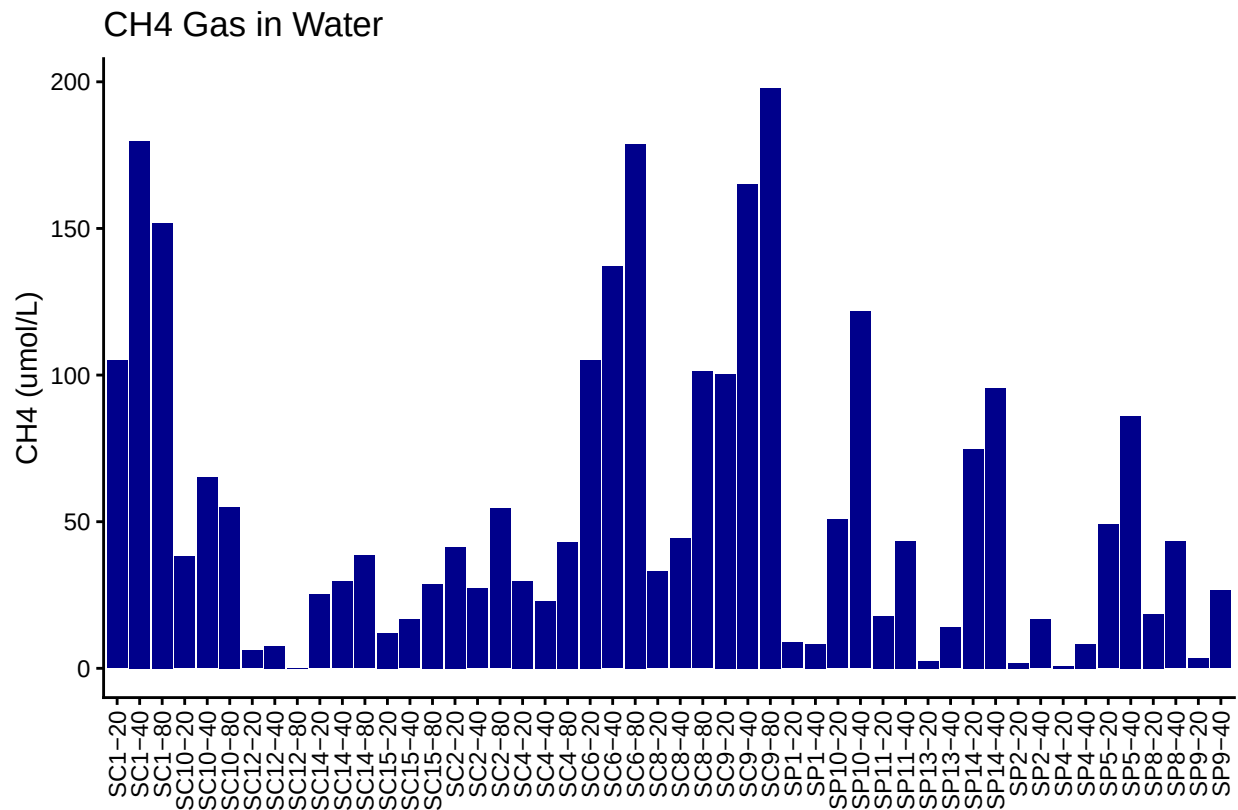
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	386.2727

5 Dilution correct samples and Check Range



6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data
```

7.1 Organize Data

7.2 Visualize Data

```
## Visualize Data
```

Figure 1 displays three panels of bar charts showing CH₄ production (uM) for aCO₂ and eCO₂ treatments at 20, 40, and 80 days. The legend indicates four treatments: eCO₂_SPPA (yellow), eCO₂_SCAM (dark green), aCO₂_SPPA (light yellow), and aCO₂_SCAM (light green).

20 Days:

- aCO₂:** eCO₂_SPPA (yellow) shows the highest production (~50 uM), followed by aCO₂_SPPA (~10 uM). eCO₂_SCAM (dark green) and aCO₂_SCAM (light green) show lower production (~10-40 uM).
- eCO₂:** eCO₂_SPPA (yellow) shows the highest production (~75 uM), followed by eCO₂_SCAM (dark green) (~100 uM). aCO₂_SPPA (light yellow) and aCO₂_SCAM (light green) show lower production (~10-25 uM).

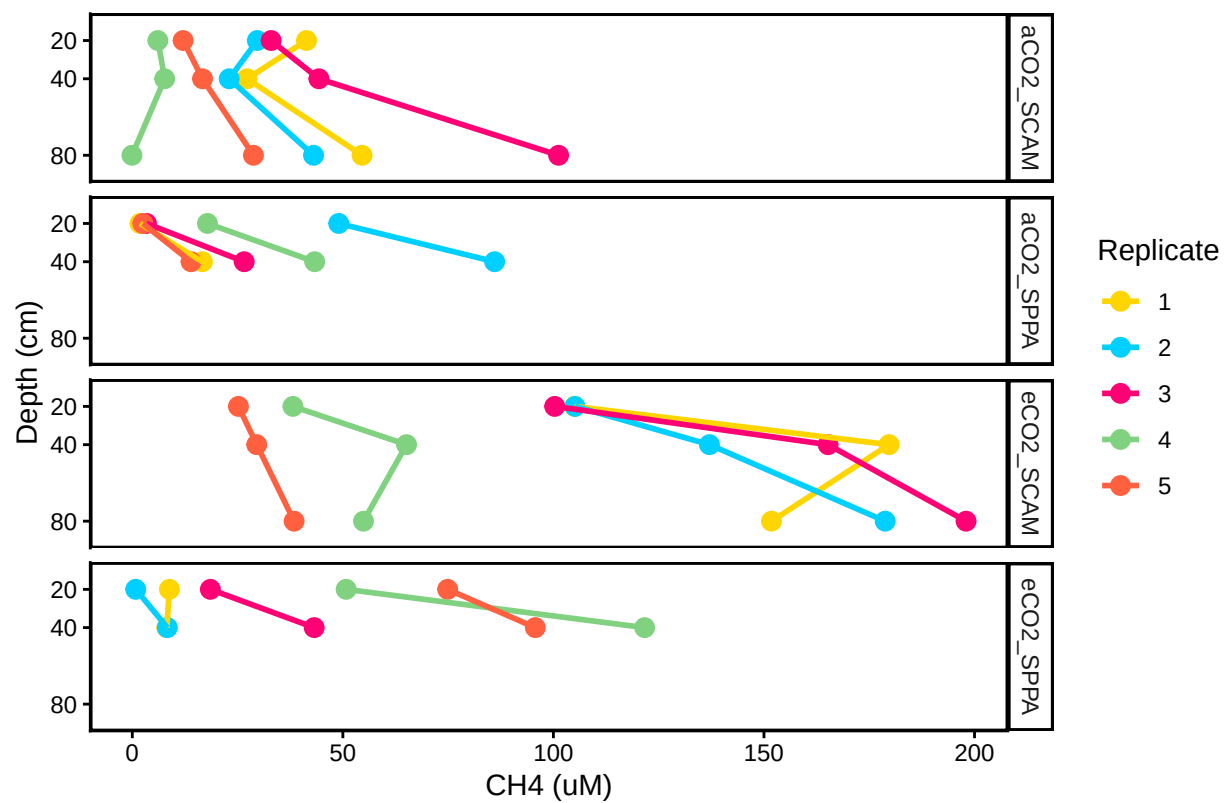
40 Days:

- aCO₂:** eCO₂_SPPA (yellow) shows the highest production (~85 uM), followed by aCO₂_SPPA (~45 uM). eCO₂_SCAM (dark green) and aCO₂_SCAM (light green) show lower production (~10-45 uM).
- eCO₂:** eCO₂_SPPA (yellow) shows the highest production (~120 uM), followed by eCO₂_SCAM (dark green) (~180 uM). aCO₂_SPPA (light yellow) and aCO₂_SCAM (light green) show lower production (~10-45 uM).

80 Days:

- aCO₂:** eCO₂_SPPA (yellow) shows the highest production (~100 uM), followed by aCO₂_SPPA (~50 uM). eCO₂_SCAM (dark green) and aCO₂_SCAM (light green) show lower production (~10-45 uM).
- eCO₂:** eCO₂_SPPA (yellow) shows the highest production (~200 uM), followed by eCO₂_SCAM (dark green) (~150 uM). aCO₂_SPPA (light yellow) and aCO₂_SCAM (light green) show lower production (~10-45 uM).

CO2xCOMM: Porewater CH4 by Replicate



#end